



ORIGINAL ARTICLE

Therapeutic Bronchoscopic Interventions for Non Foreign-Body Removal Indications in Children

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Received: 17 September 2024 | **Revised:** 13 November 2024 | **Accepted:** 19 December 2024

Funding: The authors received no specific funding for this work.

Keywords: bronchoscopy | central airway obstruction | tuberculosis | tumor

ABSTRACT

Background: The indications for pediatric airway endoscopy are expanding and a variety of therapeutic interventions are feasible for central airway obstruction (CAO) and other central airway pathologies, apart from foreign body removal.

Methods: In this retrospective chart review from four centers, we describe the indications, procedures, outcomes, and complications of therapeutic bronchoscopic interventions in children for non foreign-body removal indications.

Results: A total of 72 children (mean age:140 [60.4] months) underwent 143 therapeutic bronchoscopic airway interventions under general anesthesia, via a rigid or flexible bronchoscope or both, between January 2015 and November 2023. 67 (93%) had CAO. The most common diagnoses were posttubercular tracheobronchial stenosis (29%) followed by central airway tumor (26%) and postintubation/posttracheostomy tracheal stenosis (26%). Airway stenosis (subglottic/tracheal/bronchial) was the commonest bronchoscopic finding (43 [59.7%]) followed by endoluminal growth/mass (21 [29.2%]). The most frequently performed procedure was airway dilatation (38, 52.7%), using balloon dilators (89.5%) or rigid bronchoscope (10.5%). Endoluminal tumor debulking was performed in 21 (29%). Assistive modalities included electrocautery (44%), mechanical debulking (10%), cryotherapy (10%), or laser (4%). Airway stent insertion alone or in combination with airway dilatation or debulking was performed in 24 (33.3%) children. The airway conduit used was supraglottic airway in 19 (26%) and rigid bronchoscope in 53 (74%) children. Immediate procedural success was observed in 66 (92%) patients. Fourteen (19.4%) patients had minor procedural complications. Thirty-three (46%) children required a repeat bronchoscopic procedure.

Conclusion: Like adults, multimodality therapeutic bronchoscopic airway interventions are feasible and safe in children for CAO and other central airway indications.

1 | Introduction

Bronchoscopy is an important diagnostic tool in pediatric pulmonology. Flexible bronchoscopy in children has largely been a diagnostic procedure and the interventional aspect, including rigid bronchoscopy has primarily been restricted to airway

foreign body removal [1, 2]. Although foreign body aspiration is the main cause of central airway obstruction (CAO) in children, other conditions presenting as CAO are also encountered. These include central airway tumors, extrinsic compression of the airways by mass lesions, postinflammatory or postintubation airway strictures, airway compression by vascular

structures, and others. Interventional bronchoscopy including multimodality airway interventions are the standard of care for management of CAO in adults. However, the use of therapeutic bronchoscopic interventions for non foreign-body indications is sparsely described in pediatrics.

The major limitations to adoption of interventional bronchoscopy in children are the challenges of instrumentation with small rigid bronchoscopes, lack of availability of smaller caliber instruments, the requirement of training, need for general anesthesia, and the limited expertise in this field among pediatric pulmonologists. With the recent improvements in flexible bronchoscope design as well as pediatric compatible instrumentation with ancillary devices, which can be used via a pediatric bronchoscope, the scope of therapeutic bronchoscopic interventions in children is expanding. A variety of pediatric procedures such as lung cryobiopsy, balloon dilatation, laser ablation, electro-surgery, and airway stent placement can now be performed for various indications at a well-equipped center by an expert proceduralist [3]. These procedures can be applied to a variety of indications like postintubation or posttubercular tracheo-bronchial stenosis, benign/malignant tumors affecting the large airways, primary or secondary tracheo-bronchomalacia, tracheo-esophageal fistula, and other indications [2]. Therapeutic airway interventions via a flexible and/or rigid bronchoscope are minimally invasive, less time-consuming, provide immediate benefits, and are associated with less morbidity. Further, they may reduce or eliminate the need for complicated surgical procedures and prevent the related morbidities associated with surgery.

The number of centers performing pediatric bronchoscopy in India are limited and only a few perform interventional pediatric bronchoscopy. The data on the interventional bronchoscopy in children for non foreign-body removal indication is sparse. Therefore, herein we have collated and describe the indications, procedures, outcomes and complications of therapeutic bronchoscopic interventions performed in children for non foreign-body extraction indications at four tertiary care centers in India.

2 | Methods

This is a retrospective chart review of records collected from four centers across India (All India Institute of Medical Sciences, New Delhi; Apollo Hospitals, Bengaluru; Aster Hospital, Bengaluru and Royal Care Hospital, Coimbatore) between January 2015 till November 2023. The Institutional Ethics Committee of all the participating centers waived the requirement for informed consent. However, the standard procedural consent was obtained from all patients or their guardians before the procedure as per the institutional guidelines. The study was approved by institutional ethics committee of the All India Institute of Medical Sciences, New Delhi.

2.1 | Subjects

Consecutive children younger than 18 years of age who underwent therapeutic interventional bronchoscopic interventions with either a

flexible or rigid bronchoscopy were included in the study. Interventions including foreign body removal, whole lung lavage and diagnostic procedures such as Endobronchial Ultrasound-guided Transbronchial Needle aspiration (EBUS-TBNA) and transbronchial lung cryobiopsy (TBLC) were excluded. The following information was retrieved from the database of each institution: (i) demographic profile including age, sex, clinical diagnosis, radiological and bronchoscopic findings (ii) anesthesia and airway conduit (Laryngeal mask airway [LMA]/rigid bronchoscope) used (iii) type of bronchoscope used, (iv) interventional procedures performed, (v) immediate procedural success, and (vi) complications.

2.2 | Procedures

All subjects underwent routine investigations such as complete blood count, coagulation profile, liver and renal function tests, and viral markers (HbsAg, HCV, HIV) before the procedure. A Chest x-ray, CT scan of the thorax and diagnostic flexible bronchoscopy examination findings were available before the procedure. A preanesthetic checkup was performed for all patients. The interventional procedures were performed by an experienced bronchoscopist who has sufficient expertise in performing various therapeutic airway interventional procedures. At three centers, procedures were performed by operators who were routinely performing therapeutic bronchoscopic airway interventions in adults. All the procedures were performed under general anesthesia, administered by an experienced anesthesiologist in the bronchoscopy suite of the respective centers. Continuous monitoring of vitals—heart rate, respiratory rate, and oxygen saturation was performed during the procedure. The procedures were performed with either a flexible bronchoscope with a working channel of minimum 2 mm and/or a rigid bronchoscope selected as per the age of patient and the procedural indication. Balloon dilatation was performed using CRE balloon dilators (Boston Scientific, USA). Self-expandable metallic stents or a silicon stent were used for airway stenting procedures depending on the indication. Electrocautery, cryotherapy, argon plasma coagulation was used as and when required.

2.3 | Data Management and Statistical Analysis

The data was compiled into a Microsoft excel sheet under various headings. The statistical package Stata 14 (StataCorp, College Station, TX) was used for statistical analysis. Categorical data were represented in descriptive fashion using number (percentage). For quantitative data, median (interquartile range [IQR]) for nonnormally distributed variables and mean (SD) for normally distributed variables was used for reporting.

3 | Results

Seventy-two children (AIIMS, New Delhi- 33, Aster Hospital, Bengaluru- 17, Apollo Hospitals, Bengaluru—12, Royal Care Hospital, Coimbatore- 10) underwent 143 therapeutic bronchoscopic intervention procedures during the study period. The baseline characteristics of the subjects are summarized in

Table 1. Mean age was 140 (60) months (Range 2 months to 215 months). Thirty-three (45.8%) children were younger than 12 years. Forty-one (57%) were male. The most common presenting symptoms were breathlessness (56, 78%) and cough (44, 61%). The other presenting symptoms/complaints included failure of decannulation of tracheostomy tube (6, 8.2%), stridor (4, 5.4%), hemoptysis (3, 4.1%), and persistent oxygen/ventilator dependency in (2, 2.7%). The median (IQR) duration of symptoms was 60 months (30–150 months). A total of 67 (93%) subjects had CAO.

Radiological findings included lung collapse (28, 38.9%), tracheal or subglottic stenosis (24, 33.3%), endoluminal mass (10, 13.9%), airway esophageal fistula (4, 5.6%), hyperinflation (of the lung or a lobe) (4, 5.6%), airway stenosis (12, 16.7%), and others (fibro cavitory disease with mediastinal nodes, and broncho mediastinal fistula) in two subjects. Airway stenosis (subglottic/tracheal/bronchial) was the commonest finding on flexible bronchoscopy observed in 43 (59.7%) children. Seven of the 43 had stenosis at more than one location. Endoluminal

growth/mass was seen in 21 (29.2%). The most common pre-intervention diagnoses were posttubercular tracheobronchial stenosis (29%) followed by central airway tumor (26%) and postintubation/posttracheostomy tracheal stenosis (26%). Other diagnoses included airway esophageal fistula (6%), extrinsic compression/malacia by vascular enlargement in congenital heart disease (3%) and others (postforeign body broncho stenosis [4], respiratory papillomatosis leading to tracheal obstruction [1], idiopathic broncho stenosis [1], and congenital subglottic stenosis [1] [10%]).

The details of various interventional procedures performed are depicted in Table 2. The commonest procedure performed was airway dilatation (38, 52.7%). Balloon dilatation was the most used method and was preferred over rigid bronchoscopic dilatation. Endoluminal tumor debulking was performed in 21 (29%). The various therapeutic interventional procedures performed to treat airway stenosis is described in Table 3. Airway stent insertion alone or in combination with airway dilatation or debulking was performed in 24 (33.3%) children. The main indications for stent insertion were postintubation airway stenosis (10 [41.6%]), posttubercular stenosis (5 [20.8%]), airway esophageal fistula (4 [16.6%]), primary or secondary airway malacia (4 [16.6%]), and juvenile recurrent respiratory papillomatosis (1 [4%]). The various types of stents used are described in Table 2. Endoluminal growth/mass was observed in 21 (29%) of cases. The various methods used for endoluminal growth debulking are described in Table 3.

All the procedures were performed under general anesthesia. The airway conduit used for general anesthesia was LMA in 19

TABLE 1 | Baseline characteristics of the study participants.

Characteristics	Total, n(%)
Age (mean ± SD)	140 ± 60.4 months
< 6 years	13 (18%)
6–12 years	20 (27.7%)
> 12 years	39 (54.1%)
Male, n(%)	41 (57%)
<i>Diagnosis</i>	
Post-TB tracheobronchial stenosis	21 (29%)
Postintubation/posttracheostomy stenosis	19 (26%)
Endoluminal tumor	19 (26%)
Extrinsic compression/malacia	2 (3%)
Airway esophageal fistula	4 (6%)
Others	7 (10%)
<i>Radiological findings</i>	
Lung collapse	28 (38.9%)
Tracheal stenosis/subglottic stenosis	24 (33.3%)
Airway mass	10 (13.9%)
Airway esophageal fistula	4 (5.6%)
Hyperinflation	4 (5.6%)
Others	2 (2.8%)
<i>Flexible bronchoscopy findings</i>	
Bronchial stenosis	19 (26.4%)
Tracheal Stenosis	17 (24%)
Subglottic stenosis	7 (10%)
Endoluminal growth	21 (29.2%)
Tracheoesophageal fistula	4 (5%)
Malacia	4 (6%)

TABLE 2 | Summary of therapeutic interventional procedures performed.

Interventional procedure	Total, n(%)
Airway dilatation	28 (39%)
Endoluminal tumor debulking	20 (28%)
Airway stent	13 (18%)
Airway dilatation + stent	10 (14%)
Endoluminal tumor debulking + stent	1 (1.4%)
<i>Method of airway dilatation</i>	
Balloon dilatation	34 (47%)
Rigid bronchoscopic dilatation	4 (5.5%)
<i>Other therapies used</i>	
Cryotherapy	7 (10%)
Electrocautery	31 (44%)
Argon plasma coagulation (APC)	7 (10%)
Laser	3 (4%)
Mechanical debulking	7 (10%)
Airway stent	24 (33%)
<i>Type of stent</i>	
Self-expanding metallic stent (SEMS)	9 (39%)
Silicon	8 (35%)
T-tube	6 (26%)

(26%) and rigid bronchoscope in 53 (74%) children. Rigid bronchoscope alone was used in 13 (18%), flexible bronchoscope alone in 14 (19%), and a combination of both was used in 45 (63%) cases.

TABLE 3 | Interventional procedures performed according to the airway pathology.

Airway stenosis	43 (59.7%)
Rigid bronchoscopic dilatation	4
Balloon dilatation	34
Stent	10
Montgomery T-tube insertion	4
Electrocautery incision	16
Cryotherapy	3
Endoluminal tumor/growth	21 (29%)
Electrocautery	13
Mechanical debulking	7
Cryotherapy	4
Laser	1
APC	5
Balloon dilatation + stent	1
Airway esophageal fistula	4 (5.5%)
SEMS	4
Bronchoscopic glue instillation	2
Extrinsic compression/Airway malacia	4 (5.5%)
SEMS	2
Montgomery T-tube	2

Immediate procedural success defined as adequate airway patency not requiring tracheostomy or an open airway surgery was observed in 66 (92%) patients. The representative pre- and postprocedure images of two cases who underwent serial balloon dilatation for posttubercular bronchial stenosis and electrosurgical snaring for intraluminal tumor mass respectively are shown in Figures 1 and 2.

The complication rate was 19.4%. All the complications observed were minor and were not associated with any mortality (Table 4). Intraluminal granulation tissue was the most common delayed complication in children with stent placement. Mild to moderate airway bleeding was reported in three procedures, tracheal injury while performing mechanical debulking was reported in one procedure, and one patient had pneumothorax during bronchial dilatation. The histopathological diagnosis of airway mass included inflammatory myofibroblastic tumor (5, 23.8%), mucoepidermoid carcinoma (5, 23.8%), chondroma (1, 4.7%), carcinoid (4, 5.5%), small round cell tumor (1, 4.7%), papilloma (1, 4.7%), tubercular growth (2, 9.4%), granulation tissue (1, 4.7%), and hemangioma (1, 4.7%). Thirty-three (46%) children required a repeat procedure, and 12 (17%) children required another interventional procedure such as surgical resection of a lobe, bronchoplasty, tracheoplasty, and cardiac repair.

4 | Discussion

This is one of the largest multicenter experiences on multimodality therapeutic bronchoscopic interventions for non foreign-body removal indications (primarily CAO) in children. The results of this study show that a variety of therapeutic bronchoscopic interventions can be performed safely and effectively by experienced interventional pulmonologists, for a

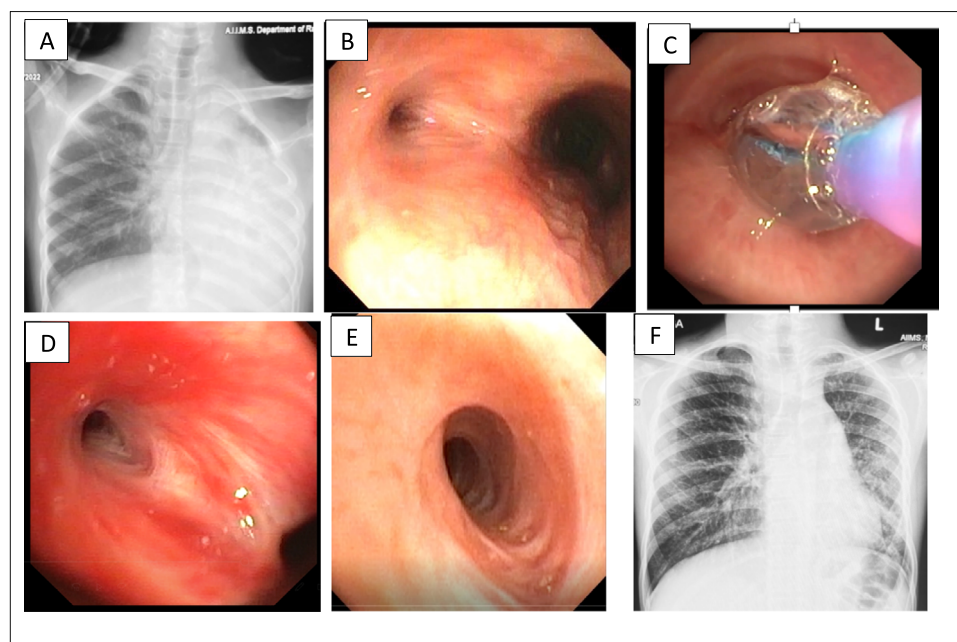


FIGURE 1 | (A) Chest radiograph at presentation suggesting collapse of left lung, (B) Flexible bronchoscopy image showing severe stenosis of left main bronchus, (C) CRE wire-guided balloon dilatation of left main bronchus, (D) Postfirst balloon dilatation of left main bronchus, (E) Significant opening of the left main bronchus postsecond balloon dilatation, (F) Postintervention chest radiograph showing significant improvement.

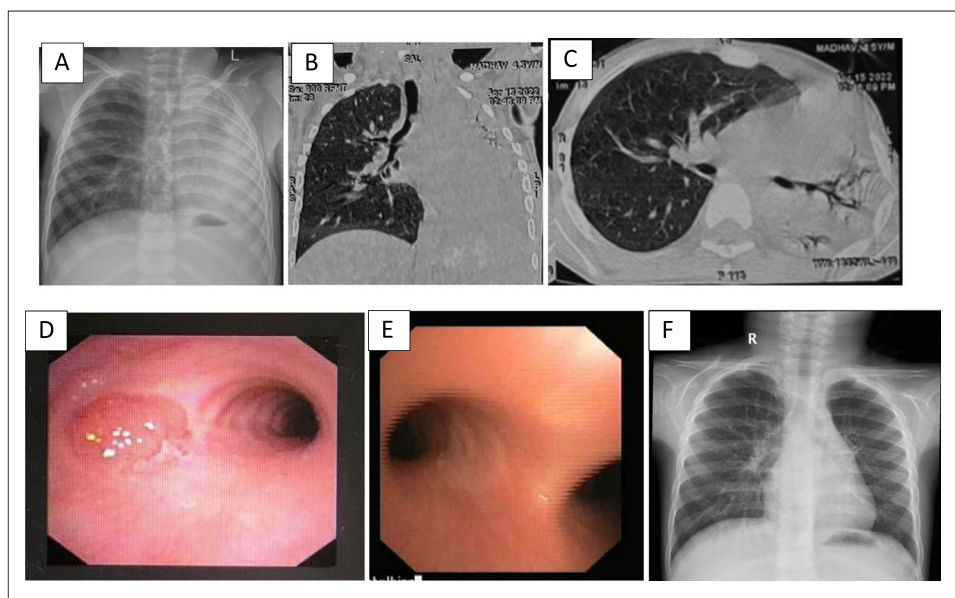


FIGURE 2 | (A) Chest radiograph showing white out left hemithorax with mediastinal shift to left (B) CT scan-nonvisualization of proximal left main bronchus (LMB) (C) distal airways patent with left lung collapse, (D) Flexible bronchoscopy image showing complete occlusion of LMB lumen by a mass. (E) Near complete patency of LMB achieved postelectrosurgical snaring of the growth. (F) Postintervention chest radiograph showing significant improvement.

TABLE 4 | Summary of complications of the interventional procedures.

Complications	14 (19.4%)
Intraluminal granulation tissue	6 (8.3%)
Airway bleeding	3 (4.1%)
Pneumothorax	1 (1.3%)
Tracheal injury	1 (1.3%)
Restenosis	2 (2.6%)
Fracture of stent	1 (1.3%)

variety of central airway pathologies via a rigid or a flexible bronchoscope. These interventions have a high rate of immediate procedural success, are safe, provide immediate symptomatic relief and are associated with minimal complications.

Tracheobronchial obstruction is rare in children but is associated with significant morbidity and mortality [4]. Tracheobronchial obstruction can be due to extrinsic compression (vascular or tumor mass), intrinsic wall abnormalities (malacia, stenosis), or due to intraluminal obstruction (malignancy, tuberculosis, benign tumor, or granulation tissue) [5]. We observed that almost 30% of patients had posttubercular tracheobronchial stenosis. The high incidence of tuberculosis in our setting, underdiagnosis of endobronchial tuberculosis, and delayed referrals could be the possible reasons for a high prevalence of TB-related CAO in our setting.

In our study, the most performed procedure was airway dilatation for airway stenosis due to various causes such as posttubercular stenosis, postintubation stenosis, or postforeign body stenosis. Airway dilatation can be performed using an expandable balloon (introduced via a flexible or a rigid

bronchoscope channel) or mechanically using serially increasing size of rigid bronchoscopes. The procedure is preferably performed under general anesthesia as it provides superior control of balloon position and optimizes patient comfort. A rigid bronchoscope as a conduit is preferred, however, the procedure can also be performed using a supraglottic airway (LMA) with a larger channel diameter flexible bronchoscope. In small children, angioplasty balloon catheters are most widely used. However, the controlled radial expansion balloons commonly used in adults can also be employed. Dilatation is initiated with the smallest balloon size chosen based on visual inspection and increasing sizes are gradually used. The balloon should be long enough to cover the entire length of the stenosis. The balloon is inflated using serially increasing pressures, and the inflation duration is usually between 30 and 60 s. Although safe, the potential adverse effects include risk of airway perforation, infection, necrosis, and delayed airway stenosis [6]. Repeated procedures are usually necessary for long-term improvement. In an observational prospective data of 59 children, who underwent 133 balloon dilatation treatments for airway stenosis, the procedural success rate was 65% [7]. In a systematic review on balloon and rigid dilatation as primary therapy for laryngo-tracheal stenosis in pediatric patients, Cheung and Chadha reported that rigid dilatation was more commonly used. Adjuvant therapies such as CO₂ or KTP laser, topical or intralesional steroid, and microdebrider was used in approximately two-thirds of the included studies. The majority of stenosis were low grade and the success rates were 50% and 53% for balloon dilatation and rigid dilatation alone, respectively. Success rates increased to 50% to 78% for balloon dilatation with adjuvant therapy and 53%–100% for rigid dilatation with adjuvant therapy. The overall morbidity and mortality rates associated with dilatation were 14% and 1%, respectively [8].

Airway stenting was the second most commonly performed procedure for various indications like airway stenosis, trachea-esophageal fistula, and airway malacia or extrinsic airway compression. The commonly used stents were SEMS (Self-expandable metallic stent), silicon stents, and Montgomery T-tube. Serio et al. described follow-up of 100 children with 235 airway stent placements. The major indications for stent placement were airway malacia and airway stenosis. One hundred and twelve silicone stents, 120 metallic stents, and three biodegradable stents were used. The complications observed were major granulation requiring stent replacement, minor granulations, stent dislocations, mucosal tear, and breakages. The immediate success rate was 97% with no fatal stent-related complications [9]. Soong et al. used flexible bronchoscopy for placement of 146 balloon-expandable metallic stents in 87 children. These stents were placed in various large airways like trachea [84], carina [15], and bronchi [47]. They noted satisfactory clinical improvements with low complication rate similar to Wang et al. [10, 11]. Stent placement can be useful for relief of severe airway malacia or stenosis when other medical and surgical options have failed or can be a bridge to more invasive surgical treatment modality. However, the possible complications related to stents with airway growth, availability of appropriate size stents and the expertise required for stent placement are major concerns in children. Biodegradable stents are likely to evolve as important modalities for management of pediatric airway stenosis in the future.

The data on therapeutic bronchoscopic procedures other than stent placement is scarce in children. Lin and Chung have described the use of various interventional bronchoscopy modalities such as mechanical debulking, electrocautery, APC, laser, and cryotherapy in management of central airway tumors [12]. Mechanical debulking is useful in intraluminal tumors. Vishwanath et al. described 30 adult patients of central airway tumors solely treated by mechanical debulking by rigid bronchoscopy, in adults. The success rate was 82.6% with complication rate of 32.3% [13].

Electrocautery, cryotherapy, APC, and laser therapy can also be used in intraluminal obstruction. Electrocautery uses high-frequency electric current leading to coagulation or tissue vaporization. It can be delivered via rigid or flexible bronchoscopy. It is easily available and cost-effective. In a retrospective series of 94 adult patients with airway obstruction, the use of electrocautery led to the luminal improvement in 78% patients with a complication rate of 6.8% [14].

APC is a noncontact technique capable of burning tissue and control bleeding. It has a penetration depth of 2–3 mm and is relatively safer. Laser therapies include CO₂ laser and Nd:YAG laser. CO₂ laser is frequently used in the larynx and upper airway because it is air-transmitted and has precise cutting effect [15]. However, in tracheobronchial trees it becomes more cumbersome. Nd:YAG laser is effective therapy for endoluminal recanalization of the trachea or main bronchi and is a popular modality in adult practice. It has a significant in-depth effect and has a potential risk of airway wall perforation, making it less suitable for infants and small children. The complication rate has been reported to be 2% and 8% for APC and laser therapy, respectively [16, 17].

The overall success rate of significant recanalization with the use of cryotherapy in central airway tumors exceeded 80% in a systematic review [18]. Cryotherapy is the use of extreme cold to destroy/adhere tissue using rapid freeze–thaw cycles. It is exclusively used for endobronchial lesions [19]. Electrocautery, cryotherapy, and APC can be used for excision of the membranous part, debulking of endoluminal mass, thus facilitating the use of balloon dilatation and stenting as observed in our study.

All the procedures in the present study were performed under general anesthesia. LMA was used as a conduit while using flexible bronchoscope alone. Both flexible and rigid bronchoscope can be used depending upon the indication, age of the patient, and therapeutic procedure to be performed. The most used flexible bronchoscope has a working channel size of minimum 2 mm to facilitate the use of smaller instruments. Rigid bronchoscope has the advantage of simultaneous ventilation during the procedure and bigger working channels. A flexible bronchoscope can also be introduced via a rigid bronchoscope for better vision and manipulation in the relatively distal airways.

Significant obstruction of the central airways is life-threatening and needs immediate treatment. Various surgical and non-surgical therapies have been used for alleviating obstruction. Surgical procedures are invasive and carry high morbidity and mortality. Nonsurgical, semi-invasive bronchoscopic modalities of treatment using flexible/rigid bronchoscope provide immediate relief of obstruction, can be done on day care basis and are associated with significant less morbidity. The use of all these procedures is well established in adults, but the data is limited regarding their use in children and adolescent age groups. Most of the case series are limited to airway stent placements and few to airway dilatation involving chiefly the subglottic area [20].

Interventional bronchoscopic procedures are less commonly performed in children due to several limitations such as small airway size, nonavailability of smaller size equipment, requirement of general anesthesia and lack of expertise. However, advanced therapeutic techniques such as laser treatment, balloon dilation, and stent placement are being performed in children with an expanding number of applications and indications in well-equipped centers by well-trained and experienced proceduralists. This is possible due to the development of new and smaller instruments and devices which include biopsy forceps, alligator forceps, cryoprobes, and the availability of approximately 4.0 mm diameter video bronchoscopes with a 2 mm working channel through which many of these instruments can be used. Another important barrier is the lack of referral until late stages in disease process, probably due to lack of pediatric interventionalists or lack of knowledge about the techniques that are possible in children. It is also noteworthy that most of these procedures were performed by adult interventional pulmonologists who routinely perform these procedures in adults. The support of adult interventional pulmonologists and a close coordination of the pediatric bronchoscopy team during the learning curve is probably a necessary step in developing this specialty within the pediatric setting.

The strengths of our study are the large sample size including data spanning over a decade from four major centers in India who are

performing therapeutic bronchoscopic interventions in children. We purposefully included only therapeutic bronchoscopic interventions for airway pathologies and excluded the foreign body removal interventions, whole lung lavage and diagnostic procedures including EBUS-TBNA, cryobiopsy, forceps biopsy, to give a clear message regarding the therapeutic modalities that are available and can be applied in children for various central airway pathologies. We also tried to broadly classify the various airway pathologies irrespective of the underlying etiology and highlight the various interventions that can be performed according to the airway pathology. The major limitation of our study is its retrospective nature. Other limitations are the lack of unified protocols across the participating centers and inability to evaluate the long-term complications, which is especially important considering the young age of the patients.

Pediatric bronchoscopy is evolving from a primarily diagnostic, semi-invasive procedure to therapeutic and diagnostic interventional bronchoscopy with the availability of “child-friendly” tools for interventions and the growing expertise in interventional bronchoscopy in the pediatric age group. These advancements have enhanced the diagnostic and therapeutic potential of this procedure, obviating the morbidity and complications of more invasive surgical procedures. Moreover, these procedures can be performed on a day care basis thus avoiding the need for hospitalization. This study shall help the pediatric physicians in better understanding of the applications of various procedures, their outcomes and complications and aid in appropriate planning and referral for the procedure.

Author Contributions

Karan Madan: conceived the idea, study design, data analysis, revised the manuscript. **Sheetal Agarwal:** study design, data collection, data analysis, prepared manuscript draft, revised the manuscript. **Jingade T. Shrikanta, Ravindra Mehta, Kaumudi Devi, Vallandramam. R. Pattabhiraman, Arjun Srinivasan, Saurabh Mittal, Vijay Hadda, Pawan Tiwari, Anant Mohan, Randeep Guleria, Kana R. Jat, Nitin Dhochak, Sushil K. Kabra:** study design, data collection, manuscript revision.

Ethics Statement

The Institutional Ethics Committee of all the participating centers waived the requirement for informed consent. However, the standard procedural consent was obtained from all patients or their guardians before the procedure as per the institutional guidelines. The study was approved by institutional ethics committee of All India Institute of Medical Sciences, New Delhi (IEC app no. AIIMS2341).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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